

Dynamic Inequality Systems Model

Complete Computational Specification: V0-V7

Document status: implementation baseline 1.0

Model class: discrete-time, group-based system dynamics model

Default time step: one year

Default simulation horizon: 50 years for intragenerational tests; 100-150 years for intergenerational tests

Primary purpose: explain and test how unequal stocks alter subsequent accumulation, opportunity, mobility, policy, and institutional dynamics.

No model element is admissible unless it has a theoretical rationale, a mathematical definition, a unit, an initialization rule, a verification test, and a calibration plan.

1. How to use this specification

This document is the implementation contract linking the research theory to the computational model. It specifies all planned phases from V0 through V7. Later empirical work may revise parameter values and, when evidence requires it, functional forms. Structural changes must be recorded as specification changes rather than introduced silently in code.

The model should be built cumulatively. Each version inherits the verified structure of the previous version and adds one coherent subsystem. A version may be promoted only when its required tests pass. The complete model is therefore not one indivisible program; it is a nested sequence of models that can be compared, disabled, and independently validated.

1.1 Authority order

1. Theoretical proposition
2. Model specification
3. Software implementation
4. Pseudo-data verification
5. Empirical calibration
6. Validation and policy analysis

When code and this specification disagree, the code is presumed wrong until the specification is deliberately amended.

1.2 Recommended first computational representation

The baseline implementation uses population groups indexed by $g = 1, \dots, G$. These may represent wealth quintiles, socioeconomic strata, regions, demographic groups, or combinations chosen for a specific empirical application. Each

group has a population share $n_g(t)$ and a vector of average or representative state variables.

The group representation is preferred initially because it preserves distributional structure while keeping stocks, flows, accounting, and feedback loops transparent. A household-level agent-based extension is possible later, but is not required to implement or test the theory.

1.3 Global conventions

- Time is discrete and indexed by t .
- Unless otherwise specified, flows are annual quantities and rates are annual rates.
- Monetary values are real currency units in a common base year.
- Stocks are measured at the start of each period; flows occur during the period; new stocks are recorded at the end.
- All probabilities remain in $[0, 1]$.
- Nonnegative stocks may not become negative through numerical error.
- Stochastic simulations must accept and record a random seed.
- Every output dataset must include model version, parameter-set identifier, scenario identifier, seed, and timestamp.

2. Core theory-to-model traceability

ID	Theoretical claim	Computational representation	First version	Required verification
T1	Inequality is both outcome and input.	Distributional measures are computed from group states and may enter opportunity, policy, legitimacy, and transition functions.	All versions; explicit feedback begins V3.	Equal-state invariance; perturbation propagation.
T2	Accumulated assets create cumulative advantage.	Assets generate capital income and expand financed investment.	V1	Disable returns; R1 must disappear.

ID	Theoretical claim	Computational representation		Required verification
			First version	
T3	Debt is not merely negative wealth.	Assets and debt are separate stocks with different rates, constraints, and risks.	V2	Equal net wealth with different gross balance sheets produces different trajectories.
T4	Credit creates asymmetric pathways.	Collateral, income, and reputation affect limits and prices; borrowing can be productive or defensive.	V2	Equalize credit terms and compare inequality.
T5	Services mediate resource conversion.	Service quality affects human-capital formation and opportunity.	V3	Equal-service counterfactual.
T6	Human capital accumulates with delays.	Education, health, and experience raise a stock that affects labor income.	V3	Impulse and delay tests.
T7	Mobility is generated by transitions.	Cohort shares move through a row-stochastic transition matrix.	V4	Rows sum to one; synthetic transition recovery.

ID	Theoretical claim	Computational representation	First version	Required verification
T8	Intergenerational persistence is multichannel.	Assets, human capital, services, networks, and debt exposure affect descendants.	V4-V5	Channel ablation tests.
T9	Networks affect information and access.	Network capital changes opportunity, job matching, and selected investment access.	V5	Disable network channel and compare.
T10	Political influence acts through policy.	Influence changes tax, transfer, regulation, or public-investment rules, never returns directly.	V5	No undeclared direct path from influence to wealth.
T11	Legitimacy and instability are conditional mediators.	Fairness affects legitimacy; low legitimacy can weaken state capacity and raise disruption only after thresholds and delays.	V6	Below-threshold invariance and recovery tests.

ID	Theoretical claim	Computational representation	First version	Required verification
T12	Balancing policies can offset reinforcing loops.	Taxes, transfers, services, insurance, debt relief, and institutional reform are explicit policy levers.	V1-V7	Policy extremes, budget closure, timing comparisons.

3. Shared state, indexing, and accounting

3.1 State vector

The complete group-level state vector is

$$\mathbf{x}_g(t) = [A_g, D_g, C_g, H_g, S_g, N_g, P_g],$$

with system-level institutional states

$$\mathbf{z}(t) = [L(t), U(t), SC(t)].$$

Where:

- A_g : assets
- D_g : debt
- C_g : credit reputation or access capacity
- H_g : human capital
- S_g : effective service quality or access
- N_g : network and social capital
- P_g : political influence
- L : institutional legitimacy
- U : instability or disruption
- SC : state capacity

Net wealth is always derived:

$$W_g(t) = A_g(t) - D_g(t).$$

3.2 Population weights and aggregate quantities

Group shares satisfy

$$\sum_g n_g(t) = 1, \quad n_g(t) \geq 0.$$

For a group variable X_g , the population-weighted mean is

$$\bar{X}(t) = \sum_g n_g(t) X_g(t).$$

Between-group dispersion may be summarized by weighted variance, ratios, top shares, or a grouped-data approximation to the Gini coefficient. The software must retain the full group distribution and must not use a single inequality statistic as a substitute for state variables.

3.3 Government budget modes

The model must support two budget modes:

Closed budget mode

$$\sum_g n_g Tax_g(t) + Rev^{other}(t) + GovBorrow(t) = \sum_g n_g Tr_g(t) + PubInv(t) + Insurance(t) + DebtService^{gov}(t).$$

Open or externally financed mode permits an explicit residual $ExternalFinance(t)$. No unexplained residual is allowed.

3.4 Period update order

Unless a version overrides it, each period follows this sequence:

1. Read beginning-of-period stocks and exogenous scenario inputs.
2. Compute aggregate and distributional indicators.
3. Determine policy rules and government budget allocations.
4. Compute labor and capital income.
5. Compute taxes, transfers, disposable resources, and consumption needs.
6. Compute credit limits, borrowing costs, borrowing demand, repayment, and default.
7. Compute investment and other stock flows.
8. Apply shocks, depreciation, and losses.
9. Update stocks simultaneously.
10. Apply cohort turnover and intergenerational transitions when active.
11. Validate bounds and accounting identities.
12. Record outputs.

Functions must use beginning-of-period values unless explicitly marked as simultaneous or lagged. This avoids accidental algebraic loops.

3.5 Master stock dictionary

Symbol	Name	Type	Unit	Bounds	Activation
A_g	Assets	stock	real currency/group member	[0, infinity)	V0 records; V1 behavioral
D_g	Debt	stock	real currency/group member	[0, infinity)	V0
C_g	Credit reputation/access capacity	stock or memory index	0-1	[0,1]	V2
H_g	Human capital	stock/index	effective labor units; default 0-1	[0,1] default	V3
S_g	Service quality/access	stock/index	0-1	[0,1]	V3
N_g	Network capital	stock/index	0-1	[0,1]	V5
P_g	Political influence	stock/share	relative influence share	[0,1], sums to 1	V5
L	Institutional legitimacy	stock/index	0-1	[0,1]	V6
U	Instability/distraction	stock/index	0-1	[0,1]	V6
SC	State capacity	stock/index	0-1	[0,1]	V6

3.6 Shared numerical rules

- Use 64-bit floating-point arithmetic at minimum.
- Default bound handling is smooth saturation in equations; clipping is a final safety assertion, not the primary behavioral mechanism.
- A flow out of a stock may not exceed the available stock in one time step.
- Debt repayment may not exceed beginning debt plus current-period interest and borrowing unless refinancing is explicitly modeled.
- Transition matrices must be normalized by a numerically stable softmax or equivalent construction.

- Missing empirical values may not be silently replaced. Imputation rules must be recorded in calibration metadata.
- The default deterministic solver is explicit annual difference equations. A smaller time-step comparison must be available for stability testing.

4. Version summary and promotion gates

Version	Subsystem added	Primary reinforcing/balancing structure	Promotion gate
V0	Accounting Sandbox	Establish exact balance-sheet, income, fiscal, and flow accounting before introducing endogenous behavioral feedback.	All accounting and reproducibility tests pass; no unexplained residuals or negative stocks.
V1	Minimal Cumulative Wealth Model	Test the central asset-accumulation mechanism and the capacity of taxes and transfers to balance it.	R1 and balancing policy loops are independently switchable, analytically understandable, and numerically stable.
V2	Credit, Debt, and Financial Fragility	Represent unequal borrowing access, borrowing price, productive and defensive borrowing, repayment, default, and asset liquidation.	Credit terms, borrowing purposes, default, and balance-sheet losses behave correctly in isolation and reduce exactly to V1 when disabled.

Version	Subsystem added	Primary reinforcing/balancing structure	Promotion gate
V3	Human Capital, Services, and Opportunity	Add slow-moving capabilities and service systems that mediate the conversion of resources into labor income and future opportunities.	Delayed stock formation, labor-income mapping, service allocation, and opportunity calculations pass isolation and reduction tests.
V4	Intergenerational Transmission and Mobility	Translate parent states into descendant initial conditions and explicit socioeconomic transition probabilities.	Cohort accounting and transition probabilities are exact; each transmission channel is independently switchable and empirically mappable.
V5	Networks and Political Allocation	Add accumulated network access and political influence that operate through explicit opportunity and policy channels.	Network and political channels are traceable, budget-consistent, symmetric under equal conditions, and contain no hidden shortcut to returns.
V6	Legitimacy, Instability, and State Capacity	Represent conditional institutional feedback in which perceived unfairness may erode legitimacy, weaken capacity, and amplify disruption.	Conditionality, threshold behavior, bounded recovery, and declared mediation paths pass; results are not presented as universal causal laws.

Version	Subsystem added	Primary reinforcing/balancing structure	Promotion gate
V7	Integrated Policy Laboratory	Provide a reproducible environment for policy packages, timing, targeting, uncertainty, fiscal accounting, and comparison across nested model versions.	The laboratory reproduces baselines, closes budgets, records complete metadata, and supports uncertainty-aware comparison without changing undeclared assumptions.

V0 - Accounting Sandbox

V0.1 Purpose

Establish exact balance-sheet, income, fiscal, and flow accounting before introducing endogenous behavioral feedback.

V0.2 Frozen scope

- Assets, debt, net wealth, labor income, exogenous capital income, taxes, transfers, consumption, saving, borrowing, repayment, asset purchases, asset losses, and government budget accounting.
- All behavioral rates are fixed inputs or scenario series.
- No credit-price feedback, human capital, services, mobility, politics, or institutional dynamics.

V0.3 Active stocks

- A_g : assets
- D_g : debt

V0.4 Auxiliary variables and controls

- $W_g = A_g - D_g$
- Y_g^L : exogenous labor income
- Y_g^K : exogenous capital income

- $Tax_g, Tr_g, Cns_g, Sav_g$

V0.5 Governing equations

$$Y_g(t) = Y_g^L(t) + Y_g^K(t) + Tr_g(t) - Tax_g(t)$$

$$Resources_g(t) = Y_g(t) + B_g(t)$$

$$Sav_g(t) = \max\{0, Resources_g(t) - Cns_g(t) - Rep_g(t)\}$$

$$A_g(t+1) = A_g(t) + AssetBuy_g(t) - AssetLoss_g(t)$$

$$D_g(t+1) = D_g(t) + B_g(t) + IntDebt_g(t) - Rep_g(t) - WriteOff_g(t)$$

$$W_g(t+1) = A_g(t+1) - D_g(t+1)$$

V0.6 Initialization specification

- Provide deterministic arrays for each group.
- Default pseudo-data: five equal-size groups with monotonic assets, non-monotonic debt, and equal labor income.
- Create a second fixture with identical groups for invariance testing.

V0.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
tau0	fixed effective tax rate	1/year	0.20
c0	fixed consumption share	dimensionless	0.75
rD0	fixed debt interest rate	1/year	0.05
loss0	asset-loss rate	1/year	0.01

V0.8 Required outputs

- All beginning and ending stocks
- Every named flow
- Income components
- Government sources and uses
- Accounting residuals

V0.9 Verification suite

- ☐ Net wealth identity holds to 10^{-10} relative tolerance.
- ☐ No flow exceeds its available source.
- ☐ Closed government budget residual is below 10^{-10} .
- ☐ Identical inputs produce bitwise-identical outputs where platform permits.
- ☐ Equal groups remain equal.
- ☐ Zero-flow fixture leaves all stocks unchanged.
- ☐ One-period hand calculation matches code.

V0.10 Calibration and empirical mapping

- No empirical calibration required.
- Use realistic magnitudes only to catch scaling errors.

V0.11 Promotion gate

All accounting and reproducibility tests pass; no unexplained residuals or negative stocks.

V1 - Minimal Cumulative Wealth Model

V1.1 Purpose

Test the central asset-accumulation mechanism and the capacity of taxes and transfers to balance it.

V1.2 Frozen scope

- Adds endogenous capital income, saving, asset investment, depreciation, heterogeneous returns, and redistribution.
- Retains exogenous labor income.
- Implements reinforcing loop R1 and balancing tax-transfer loops.

V1.3 Active stocks

- Inherited V0: A_g, D_g

V1.4 Auxiliary variables and controls

- Return rate r_g^A
- Disposable income
- Saving propensity
- Investment
- Inequality measures

V1.5 Governing equations

$$Y_g^K(t) = r_g^A(t)A_g(t)$$

$$Y_g^{disp}(t) = Y_g^L(t) + Y_g^K(t) + Tr_g(t) - Tax_g(t) - Rep_g(t)$$

$$s_g(t) = \sigma (\alpha_s + \beta_W \log(1 + \max(W_g, 0)) + \beta_Y \log(1 + \max(Y_g^{disp}, 0)))$$

$$I_g^A(t) = s_g(t) \max(Y_g^{disp}(t), 0)$$

$$A_g(t+1) = A_g(t) + I_g^A(t) + r_g^{app}(t)A_g(t) - \delta_A A_g(t) - Loss_g^{shock}(t)$$

A baseline progressive tax rule is

$$Tax_g(t) = \tau_Y(Y_g)Y_g + \tau_W(W_g) \max(W_g, 0),$$

with transfers allocated through an explicit rule that satisfies the selected government-budget mode.

V1.6 Initialization specification

- Use the V0 fixtures plus log-spaced asset holdings.
- Run equal-return and heterogeneous-return cases.
- Set debt to zero in the simplest R1 isolation run.

V1.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
rA_base	baseline capital return	1/year	0.04
rApp	asset appreciation	1/year	0.01
deltaA	asset depreciation/loss	1/year	0.01
alpha_s	saving intercept	dimensionless	-1.0
beta_W	wealth effect on saving	dimensionless	0.15
tauY	income tax schedule	dimensionless	scenario
tauW	wealth tax schedule	1/year	scenario

V1.8 Required outputs

- Wealth and asset paths by group
- Capital-income share
- Saving and investment rates
- Gini/top-share/group-gap measures
- Tax and transfer incidence
- Loop-isolation comparisons

V1.9 Verification suite

- ☐ R1 disabled: setting returns and appreciation to zero removes return-driven divergence.
- ☐ Equal-state invariance under equal parameters.
- ☐ Monotonicity: raising r_g^A weakly raises terminal assets absent offsetting policy.
- ☐ Full confiscatory return tax neutralizes net capital return in a controlled fixture.
- ☐ No-tax/no-transfer and balanced redistribution fixtures match analytic expectations.
- ☐ Time-step sensitivity using half-year steps stays within a declared 1% terminal-stock tolerance for smooth cases.

V1.10 Calibration and empirical mapping

- SCF asset stocks, income, saving, and portfolio composition.
- National accounts or asset-class return series.
- Tax and transfer incidence data.
- Pseudo-data parameter recovery before real estimation.

V1.11 Promotion gate

R1 and balancing policy loops are independently switchable, analytically understandable, and numerically stable.

V2 - Credit, Debt, and Financial Fragility

V2.1 Purpose

Represent unequal borrowing access, borrowing price, productive and defensive borrowing, repayment, default, and asset liquidation.

V2.2 Frozen scope

- Adds credit reputation/access capacity, endogenous borrowing limits and interest rates, productive and defensive credit demand, debt burden, delinquency, default, write-off, and collateral loss.
- Implements reinforcing credit loop R2 and debt/precarity loop R4.

V2.3 Active stocks

- A_g, D_g
- C_g : credit reputation/access capacity

V2.4 Auxiliary variables and controls

- Borrowing limit
- Interest rate
- Debt-service ratio
- Liquidity need
- Productive investment opportunity
- Default probability
- Collateral seizure

V2.5 Governing equations

$$Limit_g(t) = \max\{0, \lambda_0 + \lambda_A A_g + \lambda_Y Y_g + \lambda_C C_g\}$$

$$r_g^D(t) = r^{market}(t) + \rho_0 + \rho_D \frac{D_g}{Y_g + \epsilon} - \rho_A \frac{A_g}{Y_g + \epsilon} - \rho_C C_g$$

$$B_g(t) = \min\{Limit_g(t) - D_g(t), B_g^{prod}(t) + B_g^{def}(t)\}_+$$

$$DSR_g(t) = \frac{RepSched_g(t) + r_g^D(t) D_g(t)}{\max(Y_g^{gross}(t), \epsilon)}$$

$$p_g^{def}(t) = \sigma(\gamma_0 + \gamma_D DSR_g + \gamma_L LiquidityGap_g - \gamma_C C_g)$$

$$Default_g(t) = p_g^{def}(t) D_g(t)$$

$$C_g(t+1) = \text{clip}_{[0,1]} [C_g + \eta_R RepSuccess_g - \eta_M Delinquency_g - \eta_F p_g^{def}]$$

$$D_g(t+1) = D_g + B_g + r_g^D D_g - Rep_g - WriteOff_g$$

$$A_g(t+1) = A_g + I_g^{own} + I_g^{credit} + App_g - Dep_g - Liquidation_g - CollateralLoss_g.$$

V2.6 Initialization specification

- Construct households/groups with equal net wealth but different asset-debt compositions.
- Create high- and low-credit-reputation fixtures.
- Generate income shocks with known timing and magnitude.

V2.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
lambda_A	collateral contribution to limit	dimensionless	0.5
lambda_Y	income contribution to limit	years	0.5
rho_C	reputation discount	1/year	0.03
gamma_D	default sensitivity to DSR	dimensionless	5.0
eta_R	reputation gain	1/year	0.03
eta_F	reputation loss after default	1/year	0.30
recovery	lender recovery fraction	dimensionless	0.40

V2.8 Required outputs

- Credit limits and rates
- Borrowing by purpose
- Debt-service ratios
- Defaults and write-offs
- Asset liquidations
- Gross versus net balance-sheet trajectories
- Credit-loop contribution to inequality

V2.9 Verification suite

- ☐ Equal net wealth with different gross positions yields different interest and fragility outcomes.
- ☐ Borrowing never exceeds available limit.
- ☐ Default probability remains bounded.
- ☐ Repayment cannot exceed debt service due.
- ☐ Equalizing limits and rates weakens credit-driven divergence.
- ☐ A negative income shock increases defensive borrowing/default in the expected direction.

- ☐ With all credit channels disabled, results reduce to V1.
- ☐ Lender/government loss accounting closes when recovery and write-offs are included.

V2.10 Calibration and empirical mapping

- SCF and PSID balance sheets and debt payments.
- Consumer Credit Panel or equivalent administrative credit data.
- Mortgage, student-loan, and consumer-credit rate spreads.
- Default and delinquency rates by borrower characteristics.

V2.11 Promotion gate

Credit terms, borrowing purposes, default, and balance-sheet losses behave correctly in isolation and reduce exactly to V1 when disabled.

V3 - Human Capital, Services, and Opportunity

V3.1 Purpose

Add slow-moving capabilities and service systems that mediate the conversion of resources into labor income and future opportunities.

V3.2 Frozen scope

- Adds human-capital and service-quality stocks, delayed formation and depreciation, labor-income production, public and private service investment, and an opportunity index.
- Implements service-human-capital-income loop R3.
- Opportunity remains an auxiliary variable, not a stock.

V3.3 Active stocks

- Inherited financial stocks
- H_g : human capital
- S_g : effective service quality/access

V3.4 Auxiliary variables and controls

- Education/health investment
- Experience accumulation
- Labor productivity
- Labor income
- Opportunity index
- Service congestion and maintenance

V3.5 Governing equations

$$I_g^H(t) = \kappa_E \text{EduInv}_g(t)^{\alpha_E} S_g(t)^{\beta_E} + \kappa_X \text{Exp}_g(t) + \kappa_{health} \text{HealthInv}_g(t)$$

$$H_g(t+1) = \text{clip}_{[0,1]} [H_g(t) + I_g^H(t) - \delta_H H_g(t)]$$

$$I_g^S(t) = \text{PubInv}_g^S(t) + \text{PrivInv}_g^S(t) + \text{CommunityInv}_g^S(t)$$

$$S_g(t+1) = \text{clip}_{[0,1]} [S_g(t) + \kappa_S I_g^S(t) - \delta_S S_g(t) - \text{Congestion}_g(t)]$$

$$Y_g^L(t) = w_0(t) H_g(t)^{\alpha_H} \text{Match}_g(t) \text{Hours}_g(t)$$

A bounded opportunity function is

$$O_g(t) = \sigma \left(\omega_0 + \omega_W \tilde{W}_g + \omega_H H_g + \omega_S S_g + \omega_C C_g - \omega_R \text{Risk}_g \right).$$

V3.6 Initialization specification

- Generate correlated A, D, H, S rather than independent draws.
- Use one fixture with equal wealth but unequal services and another with equal services but unequal wealth.
- Apply one-period public-service impulses to test delayed effects.

V3.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
delta_H	human-capital depreciation	1/year	0.02
delta_S	service deterioration	1/year	0.03
alpha_H	labor-income elasticity	dimensionless	0.7
beta_E	service complementarity	dimensionless	0.3
omega_*	opportunity weights	dimensionless	pseudo-data normalized
delay_H	formation delay	years	1-5

V3.8 Required outputs

- Human-capital and service paths
- Labor-income decomposition
- Opportunity distributions
- Public/private service investment
- Delayed response functions
- Contribution of service equalization to inequality and mobility precursors

V3.9 Verification suite

- ☐ Human capital cannot jump before its declared delay.
- ☐ Equalizing service quality reduces the service-mediated labor-income gap.
- ☐ Zero service investment with positive depreciation lowers S .
- ☐ Higher H weakly raises labor income under fixed matching and hours.
- ☐ Opportunity remains in $[0, 1]$.
- ☐ When H/S channels are disabled, results reduce to V2.
- ☐ Dimensional consistency of all investment-to-index conversion coefficients.

V3.10 Calibration and empirical mapping

- Education attainment and expenditure; school quality and neighborhood measures.
- Health status, healthcare access, and labor-productivity evidence.
- Earnings functions using PSID/Census/ACS or country equivalents.
- Public service expenditure and quality indicators.

V3.11 Promotion gate

Delayed stock formation, labor-income mapping, service allocation, and opportunity calculations pass isolation and reduction tests.

V4 - Intergenerational Transmission and Mobility

V4.1 Purpose

Translate parent states into descendant initial conditions and explicit socioeconomic transition probabilities.

V4.2 Frozen scope

- Adds cohorts or generations, inheritance and gifts, childhood investment/exposure, cohort replacement, and a row-stochastic mobility transition matrix.
- Implements reinforcing intergenerational loop R6.

- Mobility is measured from transitions and rank changes, never stored as a stock.

V4.3 Active stocks

- All inherited group stocks by cohort
- Population shares $n_{g,c}$

V4.4 Auxiliary variables and controls

- Asset transfer
- Debt exposure
- Child human-capital endowment
- Service exposure
- Transition matrix M_{ij}
- Birth and cohort-exit rates

V4.5 Governing equations

$$Inh_{child,g}^A = \tau_A(1 - estateTax_g)A_{parent,g}$$

$$H_{child,g}^{init} = \text{clip}_{[0,1]} [h_0 + \chi_H H_{parent,g} + \chi_S S_{child,g} + \chi_I ChildInv_g]$$

Define destination scores

$$q_{ij} = \alpha_j + \beta_{jO} O_i + \beta_{jH} H_i + \beta_{jW} \tilde{W}_i + \beta_{jS} S_i + \Gamma_j Institutions,$$

then

$$M_{ij} = \frac{\exp(q_{ij})}{\sum_k \exp(q_{ik})}.$$

Cohort shares update as

$$n_j^{new}(t+1) = \sum_i n_i^{parent}(t) M_{ij}(t).$$

Initial descendant stocks are conditional expectations or sampled distributions based on parental states, transition destination, and policy.

V4.6 Initialization specification

- Create parent cohorts with known states and a known synthetic transition matrix.
- Use generation intervals such as 25 years while financial stocks continue annually.
- Support deterministic expected-value and stochastic cohort-sampling modes.

V4.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
tau_A	transfer fraction	dimensionless	0.5
estateTax	estate tax schedule	dimensionless	scenario
chi_H	parent-H persistence	dimensionless	0.3
chi_S	service exposure effect	dimensionless	0.3
beta_j*	transition-score coefficients	dimensionless	synthetic then estimated
genLength	generation interval	years	25

V4.8 Required outputs

- Transition matrices
- Absolute and relative mobility
- Rank-rank association
- Intergenerational elasticities
- Inherited asset and human-capital shares
- Channel-specific persistence contributions

V4.9 Verification suite

- ☐ Every transition probability is in $[0, 1]$ and each row sums to one within 10^{-12} .
- ☐ Population shares remain nonnegative and sum to one.
- ☐ Identity transition fixture reproduces parent distribution.
- ☐ Uniform transition fixture removes origin dependence.
- ☐ Disabling inheritance removes only asset-transfer channel.
- ☐ Synthetic transition parameters are recoverable under identified fixtures.
- ☐ Annual and generational clocks synchronize without double updating.
- ☐ V4 reduces to V3 when cohort turnover and transmission are disabled.

V4.10 Calibration and empirical mapping

- Opportunity Insights or equivalent mobility matrices.
- PSID parent-child income, wealth, and education links.
- Inheritance and inter vivos transfer data.
- Neighborhood and childhood-exposure estimates.

V4.11 Promotion gate

Cohort accounting and transition probabilities are exact; each transmission channel is independently switchable and empirically mappable.

V5 - Networks and Political Allocation

V5.1 Purpose

Add accumulated network access and political influence that operate through explicit opportunity and policy channels.

V5.2 Frozen scope

- Adds network capital, political influence, organized countervailing power, endogenous tax/transfer/public-investment parameters, and policy allocation.
- Implements network reinforcement and political-allocation loop R5.
- Political influence may not directly alter wealth or returns except through a declared institutional rule.

V5.3 Active stocks

- N_g : network capital
- P_g : political influence share

V5.4 Auxiliary variables and controls

- Job and investment matching
- Political investment
- Counter-mobilization
- Policy weights
- Endogenous tax, transfer, regulation, and service-allocation rules

V5.5 Governing equations

$$N_g(t+1) = \text{clip}_{[0,1]} [N_g + \text{Form}_g - \text{MaintLoss}_g - \text{Exclusion}_g]$$

$$Match_g(t) = \sigma(m_0 + m_H H_g + m_N N_g + m_S S_g).$$

Raw political capacity is

$$\tilde{P}_g(t+1) = (1 - \delta_P)P_g + \phi_W f(W_g) + \phi_N N_g + Org_g - Counter_g,$$

and influence shares are normalized:

$$P_g(t+1) = \frac{\max(\tilde{P}_g, 0)}{\sum_k \max(\tilde{P}_k, 0) + \epsilon}.$$

A generic policy rule is

$$Policy_k(t) = Policy_k^{base} + \psi_k \sum_g P_g(t) Preference_{gk} + DemocraticConstraint_k(t).$$

Public service allocation must satisfy

$$\sum_g PubInv_g^S(t) = PubInv(t),$$

with allocation shares generated by a transparent function of need, population, political influence, and statutory constraints.

V5.6 Initialization specification

- Use equal-network/equal-influence fixtures.
- Create wealth-different but preference-identical groups to isolate resource-to-influence mapping.
- Create influence-different groups with fixed policy preferences to isolate allocation effects.

V5.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
delta_N	network decay	1/year	0.05
delta_P	influence decay	1/year	0.10
phi_W	wealth-to-influence effect	dimensionless	scenario/estimated

Symbol	Meaning	Unit	Pseudo-data default
phi_N	network-to-influence effect	dimensionless	scenario
psi_k	policy responsiveness	policy units	context-specific
demConstraint	democratic/institutional constraint	dimensionless	0-1

V5.8 Required outputs

- Network stocks and matching effects
- Influence shares
- Policy-rule components
- Tax and transfer changes attributable to influence
- Service allocation by need versus influence
- Counterfactual outcomes with political feedback disabled

V5.9 Verification suite

- ☐ Influence shares are bounded and sum to one.
- ☐ No direct computational edge from P_g to A_g , Y_g^K , or r_g^A unless mediated by a named policy variable.
- ☐ Public allocation exactly exhausts the declared budget.
- ☐ Equal influence and preferences produce symmetric policy treatment.
- ☐ Setting policy responsiveness to zero reduces model to V4 plus passive networks.
- ☐ Network increases improve declared matching/access channels only.
- ☐ Countervailing power can offset concentrated influence in a controlled fixture.

V5.10 Calibration and empirical mapping

- Network proxies from surveys, referrals, occupational transitions, and institutional access.
- Campaign finance, lobbying, organizational membership, and policy responsiveness evidence.
- Tax incidence, regulatory, and public-spending allocation data.

V5.11 Promotion gate

Network and political channels are traceable, budget-consistent, symmetric under equal conditions, and contain no hidden shortcut to returns.

V6 - Legitimacy, Instability, and State Capacity

V6.1 Purpose

Represent conditional institutional feedback in which perceived unfairness may erode legitimacy, weaken capacity, and amplify disruption.

V6.2 Frozen scope

- Adds perceived fairness, institutional legitimacy, instability/disruption, and state capacity.
- Implements conditional loop R7 with thresholds, delays, and recovery.
- Inequality alone may not mechanically create instability. Context and mediation are mandatory.

V6.3 Active stocks

- L : legitimacy
- U : instability/disruption
- SC : state capacity

V6.4 Auxiliary variables and controls

- Perceived fairness
- Policy responsiveness gap
- Service-performance gap
- Institutional inclusion
- Escalation pressure
- Recovery effort

V6.5 Governing equations

$$Fair(t) = \sigma (f_0 - f_I Inequality(t) - f_G PolicyGap(t) - f_S ServiceGap(t) + f_M Mobility(t))$$

$$L(t+1) = \text{clip}_{[0,1]} [L + \eta_F (Fair - L) - \eta_C CorruptionShock - \eta_E Exclusion]$$

$$Pressure(t) = \max\{0, \theta_I Inequality + \theta_G PolicyGap + \theta_X Shock - \theta_L L - \Theta_U\}$$

$$U(t+1) = \text{clip}_{[0,1]} [U + \eta_U Pressure(1 - U) - \eta_R RecoveryU]$$

$$SC(t+1) = \text{clip}_{[0,1]} [SC + \eta_{rev} RevenueCapacity + \eta_{leg} L - \eta_{dis} U - \delta_{SC} SC]$$

State capacity modifies tax collection, public-investment effectiveness, service maintenance, and shock response through explicit multiplicative factors.

V6.6 Initialization specification

- Create high- and low-legitimacy institutional contexts at identical inequality.
- Construct threshold fixtures just below and above Θ_U .
- Apply temporary shocks and verify recovery dynamics.

V6.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
eta_F	legitimacy adjustment	1/year	0.10
Theta_U	instability threshold	index	0.5
eta_U	escalation speed	1/year	0.20
eta_R	recovery speed	1/year	0.10
eta_leg	legitimacy contribution to capacity	1/year	0.05
eta_dis	disruption damage to capacity	1/year	0.15

V6.8 Required outputs

- Fairness, legitimacy, instability, and capacity paths
- Threshold crossings
- Revenue-collection effectiveness
- Service-maintenance impairment
- Shock and recovery durations
- Institutional-loop contribution to inequality outcomes

V6.9 Verification suite

- ☐ High inequality below the mediated threshold does not automatically raise instability.
- ☐ Increasing legitimacy lowers escalation pressure with other inputs fixed.
- ☐ Temporary shock decays after removal when recovery is positive.
- ☐ All institutional indices remain bounded.
- ☐ State-capacity effects operate only through declared fiscal, service, and response channels.
- ☐ Setting all institutional couplings to zero reduces to V5.
- ☐ Alternative threshold specifications can be swapped without changing interfaces.

V6.10 Calibration and empirical mapping

- Survey measures of trust, fairness, legitimacy, and institutional confidence.

- Political stability, protest, conflict, and policy-volatility data.
- Tax capacity, administrative quality, and service-delivery indicators.
- Use wide priors and structural sensitivity because causal evidence is context dependent.

V6.11 Promotion gate

Conditionality, threshold behavior, bounded recovery, and declared mediation paths pass; results are not presented as universal causal laws.

V7 - Integrated Policy Laboratory

V7.1 Purpose

Provide a reproducible environment for policy packages, timing, targeting, uncertainty, fiscal accounting, and comparison across nested model versions.

V7.2 Frozen scope

- Includes all verified subsystems and scenario controls.
- Adds no new causal theory by default; it exposes policy levers and experimental protocols.
- Supports reduced-model comparison to determine which subsystems drive each result.

V7.3 Active stocks

- All V0-V6 states

V7.4 Auxiliary variables and controls

- Scenario schedule
- Eligibility and targeting
- Implementation delay
- Administrative cost
- Compliance and take-up
- Policy financing
- Outcome and welfare metrics

V7.5 Governing equations

Each policy p is represented by a schedule

$$Policy_p(t) = Magnitude_p \times Ramp_p(t; t_0, duration, phaseout) \times Coverage_p(t) \times Effectiveness_p(t).$$

Fiscal cost is

$$Cost_p(t) = DirectTransfer_p + PublicInvestment_p + Administration_p + CreditLoss_p - Revenue_p.$$

A generic multi-target calibration or policy loss may be

$$\mathcal{L}(\theta) = \sum_m w_m \left(\frac{Moment_m^{sim}(\theta) - Moment_m^{obs}}{Scale_m} \right)^2.$$

Policy comparisons report distributions, not only means, and include uncertainty intervals from parameter, initialization, stochastic, and structural uncertainty.

V7.6 Initialization specification

- Use a named calibrated baseline plus a pseudo-data baseline.
- Every scenario inherits a frozen parameter set except for declared policy levers.
- Run paired seeds for stochastic comparisons.

V7.7 Minimum parameter catalog

Symbol	Meaning	Unit	Pseudo-data default
start_p	policy start year	year	scenario
duration_p	policy duration	years	scenario
coverage_p	eligible population share	dimensionless	scenario
takeup_p	participation rate	dimensionless	scenario
admin_p	administrative cost share	dimensionless	scenario
finance_p	tax/borrowing/external mix	shares	scenario

V7.8 Required outputs

- Full state and flow trajectories
- Distributional outcomes and mobility
- Fiscal costs and sources
- Policy incidence
- Loop-contribution and ablation results
- Uncertainty intervals
- Short-, medium-, and long-run effects
- Scenario metadata sufficient for exact reproduction

V7.9 Verification suite

- ☐ Same baseline, parameters, scenario, and seed reproduce identical results.
- ☐ Paired-policy comparison changes only declared levers.
- ☐ All fiscal sources and uses reconcile.
- ☐ Policy schedules activate and phase out at exact requested dates.
- ☐ Equivalent present-value spending with different timing can be compared.
- ☐ Nested-version comparisons run from the same initial conditions.
- ☐ Global sensitivity and uncertainty routines recover known pseudo-data rankings.
- ☐ No policy conclusion is accepted unless robust across a declared uncertainty set and relevant structural alternatives.

V7.10 Calibration and empirical mapping

- Use the V0-V6 parameter catalog and empirical targets.
- Estimate implementation, take-up, administrative cost, and policy-effect parameters from program evaluations.
- Reserve a holdout period or outcome set for validation.

V7.11 Promotion gate

The laboratory reproduces baselines, closes budgets, records complete metadata, and supports uncertainty-aware comparison without changing undeclared assumptions.

5. Complete variable and flow dictionary

The following dictionary defines the intended complete model interface. Specific empirical applications may add disaggregated variables, but they may not silently change the meaning of these core concepts.

Symbol	Meaning	Class	Unit	Activation	Inflow/driven by	Outflow/use
A_g	Assets	stock	currency/person	V0/V1	Asset purchases, appreciation, inheritance	Depreciation, liquidation, collateral loss
D_g	Debt	stock	currency/person	V0/V2	Borrowing, accrued interest	Repayment, write-off

Symbol	Meaning	Class	Unit	Activation	Inflow/drive	Outflow/use
W_g	Net wealth	auxiliary	currency/person	V0	$A_g - D_g$	Derived only
Y^L_g	Labor income	flow/auxiliary	currency/person/year	V0; exogenous; V3	Human capital, matching, hours	Taxes do not change gross value
Y^K_g	Capital income	flow	currency/person/year	V0; exogenous; V1	Assets and returns	Taxation after gross calculation
I^A_g	Asset investment	flow	currency/person/year	V0	Saving and productive credit	Adds to assets
B_g	Borrowing	flow	currency/person/year	V0; V2	Credit demand and limits	Adds to debt
Rep_g	Repayment	flow	currency/person/year	V0; V2	Resources and schedule	Reduces debt
Default_g	Default	event/flow	currency/person/year	V2	Debt burden and shocks	Write-off and collateral effects
C_g	Credit reputation	stock	index	V2	Successful repayment	Delinquency/default
H_g	Human capital	stock	effective labor index	V3	Education, health, experience	Depreciation/obsolescence
S_g	Service quality/access	stock	index	V3	Public/private investment	Deterioration/congestion
O_g	Opportunity	auxiliary	probability-like index	V3	Stocks and institutions	Used in transitions/decisions

Symbol	Meaning	Class	Unit	Activation	Inflow/drive	Outflow/use
M_ij	Mobility transition	probability	probability/year or generation	V4	Origin state and institutions	Rows sum to one
N_g	Network capital	stock	index	V5	Formation and maintenance	Decay and exclusion
P_g	Political influence	stock/share	share	V5	Resources, networks, organization	Decay/counter-mobilization
Tax_g	Taxes	flow	currency/person/year	V6 onward	Policy rules and bases	Government revenue
Tr_g	Transfers	flow	currency/person/year	V6 onward	Policy eligibility/allocation	Household resource flow
L	Legitimacy	stock	index	V6	Fairness and performance	Exclusion/corruption
U	Instability	stock	index	V6	Escalation pressure	Recovery
SC	State capacity	stock	index	V6	Revenue, legitimacy, investment	Disruption, depreciation

6. Parameter governance

Every parameter record must contain:

- symbol and plain-language definition;
- equation and subsystem;
- unit and valid range;
- sign expectation;
- pseudo-data default;
- empirical source or prior;

- estimation/calibration method;
- uncertainty distribution;
- correlation with other parameters;
- version first activated;
- sensitivity ranking;
- change history.

Parameters must be stored outside source code in versioned machine-readable files such as YAML, JSON, or Parquet. The simulator may not contain unexplained numeric literals other than numerical tolerances.

6.1 Parameter classes

1. **Accounting parameters:** conversion factors and timing conventions.
2. **Behavioral parameters:** saving, borrowing, repayment, investment, and matching responses.
3. **Technological parameters:** production and accumulation elasticities.
4. **Institutional parameters:** tax schedules, policy responsiveness, state capacity, and constraints.
5. **Transition parameters:** mobility and intergenerational persistence.
6. **Shock parameters:** frequency, magnitude, duration, and covariance.
7. **Measurement parameters:** mappings from observed variables to latent indices.

7. Pseudo-data design

Pseudo data serve three different purposes and should be stored separately.

7.1 Deterministic fixtures

Small arrays with hand-computable results for accounting, symmetry, bounds, and edge cases. These are unit-test data, not simulated populations.

7.2 Synthetic populations

Generate correlated group or household attributes using a latent rank variable z :

$$\log(A + 1) = a_0 + a_1 z + \epsilon_A,$$

$$D = d_0 + d_1 \max(-z, 0) + d_2 A + \epsilon_D,$$

$$H = \sigma(h_0 + h_1 z + \epsilon_H), \quad S = \sigma(s_0 + s_1 z + \epsilon_S).$$

The covariance matrix and random seed must be saved. Independent draws are prohibited for the main synthetic baseline because they erase theoretically important correlations.

7.3 Synthetic truth datasets

Run the model with known parameters, add controlled measurement error, and then apply the intended estimation pipeline. These datasets test identifiability and parameter recovery.

8. Calibration specification

8.1 Calibration hierarchy

1. Directly observed initial states.
2. Parameters estimated from microdata or quasi-experimental evidence.
3. Parameters inferred by matching dynamic moments.
4. Literature-informed priors.
5. Explicitly uncertain structural parameters explored through scenarios.

8.2 Target moments

The full target set may include:

- asset and debt distributions;
- gross and net balance-sheet positions;
- labor and capital income distributions;
- saving and investment rates;
- borrowing rates, interest spreads, debt service, delinquency, and default;
- human-capital, service, and earnings gradients;
- intergenerational transition matrices and rank-rank relationships;
- tax and transfer incidence;
- public-service allocation;
- political influence proxies and policy responsiveness;
- legitimacy, capacity, and disruption indicators.

8.3 Objective function

A normalized weighted-moment loss is

$$\mathcal{L}(\theta) = \sum_{m=1}^M w_m \left(\frac{\mu_m^{sim}(\theta) - \mu_m^{obs}}{s_m} \right)^2,$$

where s_m is a scale such as the observed standard error or a substantively meaningful tolerance. Bayesian calibration may replace or supplement optimization. Weights and scales must be declared before viewing policy results.

8.4 Identification requirements

- Perform synthetic parameter recovery for every proposed estimator.
- Report posterior or sampling correlations among parameters.
- Profile weakly identified parameters.
- Compare reduced and expanded models.
- Do not interpret a calibrated parameter causally unless supported by identification outside the model.

9. Validation and robustness

Validation is layered:

1. **Structural validation:** equations and boundaries are theoretically plausible.
2. **Dimensional validation:** every equation is unit-consistent.
3. **Code verification:** implementation matches equations.
4. **Behavior reproduction:** model matches multiple historical patterns simultaneously.
5. **Holdout validation:** fit is tested on unused periods, cohorts, regions, or outcomes.
6. **Sensitivity analysis:** local and global sensitivity are reported.
7. **Structural uncertainty:** alternative functional forms and boundaries are compared.
8. **Policy robustness:** conclusions persist across plausible parameter and structural sets.

Minimum global methods should include Morris screening or equivalent for high-dimensional triage and Sobol or variance-based indices for the smaller influential set. Parameter uncertainty, initial-state uncertainty, stochastic uncertainty, and model-structure uncertainty must be distinguished.

10. Software architecture

A recommended repository is:

```
inequality-model/  
  pyproject.toml  
  README.md  
  configs/  
    parameters/  
    scenarios/  
    calibration/  
  data/  
    raw/  
    interim/
```

```
processed/
synthetic/
src/inequality_model/
state.py
accounting.py
v0_accounting.py
v1_wealth.py
v2_credit.py
v3_human_services.py
v4_generations.py
v5_networks_politics.py
v6_institutions.py
v7_policy.py
simulator.py
metrics.py
calibration.py
validation.py
io.py
tests/
  fixtures/
  test_accounting.py
  test_bounds.py
  test_reduction.py
  test_transitions.py
  test_recovery.py
notebooks/
reports/
docs/
```

10.1 Required interfaces

Each subsystem should expose:

- `validate_inputs(state, params)`
- `compute_auxiliaries(state, params, exogenous)`
- `compute_flows(state, auxiliaries, params)`
- `update_state(state, flows, dt)`
- `validate_outputs(previous_state, new_state, flows)`

The top-level simulator activates subsystems through a version or feature configuration. A V7 run must be able to disable later subsystems and exactly reproduce lower-version behavior.

10.2 Output schema

At minimum each long-format output record contains:

- run ID;
- model version;
- scenario ID;
- parameter-set ID;
- seed;
- time;
- group/cohort;
- variable;
- value;
- unit.

A separate run manifest stores git commit, software environment, input hashes, and configuration files.

11. Master verification matrix

Test ID	Version	Requirement	Mode	Criterion
V0-01	V0	Net wealth identity holds to 10^{-10} relative tolerance.	Automated	Must pass before promotion
V0-02	V0	No flow exceeds its available source.	Automated	Must pass before promotion
V0-03	V0	Closed government budget residual is below 10^{-10} .	Automated	Must pass before promotion
V0-04	V0	Identical inputs produce bitwise-identical outputs where platform permits.	Automated	Must pass before promotion
V0-05	V0	Equal groups remain equal.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V0-06	V0	Zero-flow fixture leaves all stocks unchanged.	Automated	Must pass before promotion
V0-07	V0	One-period hand calculation matches code.	Automated	Must pass before promotion
V1-01	V1	R1 disabled: setting returns and appreciation to zero removes return-driven divergence.	Automated	Must pass before promotion
V1-02	V1	Equal-state invariance under equal parameters.	Automated	Must pass before promotion
V1-03	V1	Monotonicity: raising r_g^A weakly raises terminal assets absent offsetting policy.	Automated	Must pass before promotion
V1-04	V1	Full confiscatory return tax neutralizes net capital return in a controlled fixture.	Automated	Must pass before promotion
V1-05	V1	No-tax/no-transfer and balanced re-distribution fixtures match analytic expectations.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V1-06	V1	Time-step sensitivity using half-year steps stays within a declared 1% terminal-stock tolerance for smooth cases.	Automated	Must pass before promotion
V2-01	V2	Equal net wealth with different gross positions yields different interest and fragility outcomes.	Automated	Must pass before promotion
V2-02	V2	Borrowing never exceeds available limit.	Automated	Must pass before promotion
V2-03	V2	Default probability remains bounded.	Automated	Must pass before promotion
V2-04	V2	Repayment cannot exceed debt service due.	Automated	Must pass before promotion
V2-05	V2	Equalizing limits and rates weakens credit-driven divergence.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V2-06	V2	A negative income shock increases defensive borrowing/default in the expected direction.	Automated	Must pass before promotion
V2-07	V2	With all credit channels disabled, results reduce to V1.	Automated	Must pass before promotion
V2-08	V2	Lender/government loss accounting closes when recovery and write-offs are included.	Automated	Must pass before promotion
V3-01	V3	Human capital cannot jump before its declared delay.	Automated	Must pass before promotion
V3-02	V3	Equalizing service quality reduces the service-mediated labor-income gap.	Automated	Must pass before promotion
V3-03	V3	Zero service investment with positive depreciation lowers S .	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V3-04	V3	Higher H weakly raises labor income under fixed matching and hours.	Automated	Must pass before promotion
V3-05	V3	Opportunity remains in $[0, 1]$.	Automated	Must pass before promotion
V3-06	V3	When H/S channels are disabled, results reduce to V2.	Automated	Must pass before promotion
V3-07	V3	Dimensional consistency of all investment-to-index conversion coefficients.	Automated	Must pass before promotion
V4-01	V4	Every transition probability is in $[0, 1]$ and each row sums to one within 10^{-12} .	Automated	Must pass before promotion
V4-02	V4	Population shares remain nonnegative and sum to one.	Automated	Must pass before promotion
V4-03	V4	Identity transition fixture reproduces parent distribution.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V4-04	V4	Uniform transition fixture removes origin dependence.	Automated	Must pass before promotion
V4-05	V4	Disabling inheritance removes only asset-transfer channel.	Automated	Must pass before promotion
V4-06	V4	Synthetic transition parameters are recoverable under identified fixtures.	Automated	Must pass before promotion
V4-07	V4	Annual and generational clocks synchronize without double updating.	Automated	Must pass before promotion
V4-08	V4	V4 reduces to V3 when cohort turnover and transmission are disabled.	Automated	Must pass before promotion
V5-01	V5	Influence shares are bounded and sum to one.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V5-02	V5	No direct computational edge from P_g to A_g , Y_g^K , or r_g^A unless mediated by a named policy variable.	Automated	Must pass before promotion
V5-03	V5	Public allocation exactly exhausts the declared budget.	Automated	Must pass before promotion
V5-04	V5	Equal influence and preferences produce symmetric policy treatment.	Automated	Must pass before promotion
V5-05	V5	Setting policy responsiveness to zero reduces model to V4 plus passive networks.	Automated	Must pass before promotion
V5-06	V5	Network increases improve declared match-ing/access channels only.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V5-07	V5	Countervailing power can offset concentrated influence in a controlled fixture.	Automated	Must pass before promotion
V6-01	V6	High inequality below the mediated threshold does not automatically raise instability.	Automated	Must pass before promotion
V6-02	V6	Increasing legitimacy lowers escalation pressure with other inputs fixed.	Automated	Must pass before promotion
V6-03	V6	Temporary shock decays after removal when recovery is positive.	Automated	Must pass before promotion
V6-04	V6	All institutional indices remain bounded.	Automated	Must pass before promotion
V6-05	V6	State-capacity effects operate only through declared fiscal, service, and response channels.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V6-06	V6	Setting all institutional couplings to zero reduces to V5.	Automated	Must pass before promotion
V6-07	V6	Alternative threshold specifications can be swapped without changing interfaces.	Automated	Must pass before promotion
V7-01	V7	Same baseline, parameters, scenario, and seed reproduce identical results.	Automated	Must pass before promotion
V7-02	V7	Paired-policy comparison changes only declared levers.	Automated	Must pass before promotion
V7-03	V7	All fiscal sources and uses reconcile.	Automated	Must pass before promotion
V7-04	V7	Policy schedules activate and phase out at exact requested dates.	Automated	Must pass before promotion
V7-05	V7	Equivalent present-value spending with different timing can be compared.	Automated	Must pass before promotion

Test ID	Version	Requirement	Mode	Criterion
V7-06	V7	Nested-version comparisons run from the same initial conditions.	Automated	Must pass before promotion
V7-07	V7	Global sensitivity and uncertainty routines recover known pseudo-data rankings.	Automated	Must pass before promotion
V7-08	V7	No policy conclusion is accepted unless robust across a declared uncertainty set and relevant structural alternatives.	Automated	Must pass before promotion

12. Empirical data crosswalk

The exact data sources depend on the selected country and population. For a U.S. implementation, the initial crosswalk is:

Construct	Candidate source	Versions	Use
Assets, debt, income, portfolio	Survey of Consumer Finances	V0-V2	Initial distributions, returns, saving, leverage
Longitudinal family outcomes	Panel Study of Income Dynamics	V1-V4	Flows, shocks, parent-child links

Construct	Candidate source	Versions	Use
Mobility and place	Opportunity Insights	V3-V4	Transition and geographic targets
Population, earnings, education	Census/ACS/CPS	V3-V4	Labor and demographic calibration
Macro rates and growth	FRED/National Accounts	V0-V3	Exogenous macro series
Tax data	IRS and policy microsimulation sources	V1/V5/V7	Tax bases and incidence
Credit histories	Consumer Credit Panel or equivalent	V2	Limits, rates, delinquency, default
Education and health services	Administrative and survey sources	V3	Service and human-capital mappings
Campaign/lobbying/finance	Politics, lobbying, and public-finance data	V5	Influence and policy allocation
Trust and institutions	General Social Survey, World Values Survey, governance data	V6	Legitimacy and capacity
Conflict/protest/stability	Elitist and governance datasets	V6	Instability and recovery

13. Implementation sequence and completion definition

13.1 Recommended sequence

1. Implement common state containers, units, configuration, and output schema.
2. Implement V0 and its deterministic fixtures.
3. Implement V1 and analytic loop-isolation experiments.
4. Implement V2 with gross-balance-sheet fixtures and shocks.
5. Implement V3 with delayed stock formation and correlated pseudo data.
6. Implement V4 with separate annual and generational clocks.
7. Implement V5 with explicit graph-level checks preventing direct political shortcuts.

8. Implement V6 as an optional conditional module with alternative threshold specifications.
9. Implement V7 scenario, calibration, uncertainty, and reporting infrastructure.

13.2 Definition of complete implementation

The entire V0-V7 model is considered implemented only when:

- every variable in the master dictionary is represented or explicitly deferred;
- every governing equation has a tested software implementation;
- every version can run independently;
- lower-version reduction tests pass from the V7 codebase;
- pseudo-data verification and synthetic recovery pass;
- all monetary and population accounting closes;
- every result is reproducible from a run manifest;
- calibration targets and uncertainty are recorded;
- policy experiments change only declared levers;
- documentation and code use consistent notation.

13.3 Change-control rule

Any change to a stock, flow, equation, timing convention, unit, boundary, or policy channel requires:

1. a specification change note;
2. an updated theory-to-model traceability row;
3. an updated variable or parameter dictionary;
4. at least one new or modified verification test;
5. a versioned migration note for existing calibration files.

Appendix A - Version dependency map

V0 Accounting

- > V1 Wealth accumulation and redistribution
 - > V2 Credit, debt, default, fragility
 - > V3 Human capital, services, opportunity
 - > V4 Intergenerational transmission and mobility
 - > V5 Networks and political allocation
 - > V6 Legitimacy, instability, state capacity
 - > V7 Integrated policy laboratory

Appendix B - Minimum scenario library

- Equal initial conditions
- Unequal assets only
- Unequal debt only
- Equal net wealth, unequal gross balance sheets
- Equal credit terms
- Equal service quality
- No inheritance
- Universal high-quality services
- Progressive tax and transfer
- Debt relief and insurance
- Political feedback disabled
- Strong democratic constraints
- Low- and high-legitimacy contexts
- Temporary macro shock
- Permanent productivity shock
- Early versus late policy with equal present-value cost
- Single policy versus coordinated package

Appendix C - Open specification decisions before empirical calibration

The model is fully specified at the architectural level, but each empirical application must freeze the following choices in its calibration configuration:

- country, period, population, and group definition;
- monetary base year and deflator;
- number and boundaries of groups;
- exact labor-income and return functional forms;
- service-index construction;
- opportunity-index construction and interpretation;
- generation interval and cohort mapping;
- policy preference representation;
- legitimacy and instability measurement model;
- government financing mode;
- calibration targets, weights, priors, and holdout design.

These are application-level choices rather than omissions from the V0-V7 architecture. They must be decided before a calibrated model is described as an empirical representation of a particular society.